



Lab Communication and Process

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7/1/2026

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97

Lab Communication



One key to a successful indirect restorations is the communication that you have with your laboratory technician. The better you can communicate your wishes to them the more likely you will get back what you want from them.





Doctor and Patient Information



SACRAMENTO/NEWPORT BEACH DELIVERY ADDRESS
700 N. SUNRISE AVE
ROSEVILLE, CA 95661
916.786.6740

DR. Joel Whiteman LICENSE # 85631
SIGNATURE: _____ PHONE: 916-446-0203
ADDRESS: 3171 Riverside Blvd CITY: Sacramento STATE: Ca ZIP: 95818
PATIENT'S NAME: Joe Shmoe AGE: 50 SEX: M
TODAY'S DATE: 2/6/23 DUE DATE*: 2/28/23

BY SIGNING THIS RX, I AGREE TO THE TERMS ON THE BACK

*DETERMINE DELIVERY DATE: SCHEDULE YOUR PATIENT AT LEAST 14 DAYS FROM TODAY. DATE YOU ARE MAILING THIS CASE + 14 DAYS = DELIVERY DATE

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10
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The first step is for you or your assistant to fill out the doctor and patient information. Labs will want a due date for the restoration. Typically, they will tell you the number of working days that they will need for a specific type of restoration. Using this information, you or your staff will give a date when the restoration should be returned. This information will also allow you to set a date for the patient's return appointment for seating. I highly recommend that a patient be scheduled for their seating appointment before they leave your office. This will help to ensure that the patient returns in a timely manner for the seating of their crown.



Type of Restoration

1. RX		TOOTH#
ALL CERAMIC	EMPRESS	_____
	AUTHENTIC	_____
	EMAX EPRESS	_____ #9 _____
	(LITHIUM DISILICATE)	_____
	ZIRCONIA	_____
PORCELAIN TO METAL	FELDSPATHIC	_____
	P.F.O.	_____
	P.F.O. HIGH NOBLE	_____
CERAMICS COMPOSITES	PFM	_____
	SINFONY	_____
ALL METAL	GOLD HIGH NOBLE	_____
	GOLD NOBLE	_____

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The next step is to indicate the type of crown or restoration that you want. Some labs will have a selection of materials others you will have to write down what you want used.



Special Instructions

2. SHADE OF PREPARATION

PREP SHADE OF TEETH #S #9: A3 ND
 PREP SHADE OF TEETH #S ND
 PREP SHADE OF TEETH #S ND

3. LENGTH OF CENTRALS TO SOFT TISSUE ZENITH

#8 #9
 SPECIAL LENGTH INSTRUCTIONS

4. SMILE DESIGN

☐ SMILE DESIGN CATALOG NAME
☐ SMILE DESIGN PAGE
☐ SMILE DESIGN STYLE
☐ MATCH PHOTOS, MAGAZINE, ETC.
☐ FOLLOW WAX UP
☐ FOLLOW TEMPS/MOCKUP
☐ COMMENTS

5. GOAL OF NEW SMILE DESIGN

☐ CHANGE SHAPE (CONTOUR)	☐ FEMINIZE SHAPE
☐ ADJUST MIDLINE	☐ YOUTHEN SMILE
☐ WIDEN BUCCAL CORRIDOR	☐ LENGTHEN TEETH
	☐ CHANGE SHADE
	☐ CLOSE DIASTEMA

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Depending on how customized you want your crown or if you are doing an anterior crown or veneer you will want to provide special instructions. These can include the shade of tooth after it has been prepared. The final length of your central incisors or a smile that the patient wants.



Special Instructions

6. INCISAL TRANSLUCENCY

☒ MINIMAL .5
☐ MODEST 1.0
☐ MAX 1.5

☐ SMILE DESIGN CATALOG
☐ SMILE DESIGN PAGE
☐ SMILE DESIGN STYLE

7. RIDGE RELIEF

☐ YES ☐ NO

OVATE MM FULL LAP BUCCAL LAP SANITARY CONTACT SANITARY SPACED

8. SURFACE TEXTURE

☐ HIGH
☒ MEDIUM
☐ LIGHT
☐ SMOOTH (NO TEXTURE)

☐ SMILE DESIGN CATALOG
☐ SMILE DESIGN PAGE
☐ SMILE DESIGN STYLE

9. SURFACE FINISH

☒ HIGH GLAZE
☐ POLISHED GLASS
☐ SATIN FINISH
☐ LOW GLOSS

☐ SMILE DESIGN CATALOG
☐ SMILE DESIGN PAGE
☐ SMILE DESIGN STYLE

10. LABORATORY CHECKLIST FOR DOCTOR'S OFFICE

☐ SHADE OF PREPARATION
☐ LENGTH OF CENTRALS TO SOFT TISSUE
☐ SMILE DESIGNS
☐ BITE REGISTRATION W/O STICK
☐ BITE REGISTRATION W/STICK
☐ FACEBOW
☐ WORKING IMPRESSIONS
☐ OPPOSING IMPRESSION/ MODEL
☐ TEMPORARY IMPRESSION/ MODEL

☐ FINAL SHADE
☐ IMPRESSION OF COPING
☐ IMPLANT ANALOGS
☐ IMPLANT ABUTMENTS

PLEASE SEND
☐ MAILING BOXES
☐ RX FORMS
☐ MAILING LABELS

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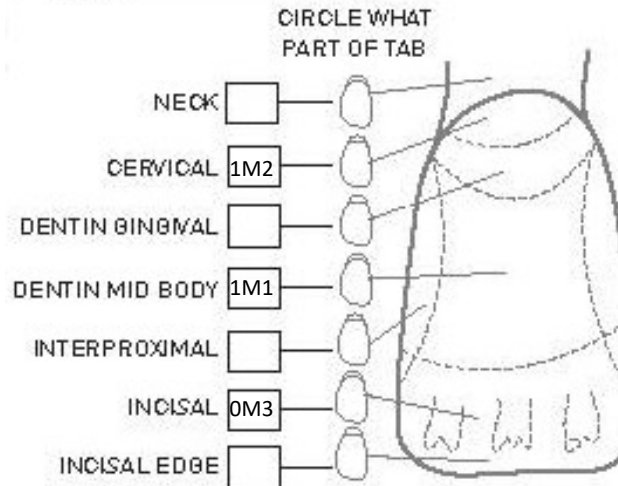
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Additionally, you can dictate to the lab the amount of translucency that you want in the incisal, the pontic shape and textures. This is not a must for all crowns but the more information that you can give your lab technician

Shade

SHADE GUIDE

TAB # See detailed shade



SHADE NOTES _____

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There are many shade guides that are available. When taking a shade there are times you will need to take a detailed shade that incorporates different portions of the tooth that is demonstrated in the picture in the slide. For this case we are using the Vita 3D master shade guide. With anterior teeth you will more often than not be giving a more detailed shade. Taking shade picture along with the prescription can be very helpful for your lab. We will go more into this in our discussion of this in the aesthetic treatment planning lecture. For posterior teeth, it is more common to use a solitary shade since it is not in the aesthetic zone.

Shade Picture



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For these pictures you will want to use a macro lens with an f stop of about 29. It is important to use a flash diffuser so that you do not get sharp flash points in the image. A slightly darker image is better, it can always be lightened with software such as Photoshop. The shade tab should be in the same vertical and horizontal plane as the tooth to give the same reflection of light. If you want multiple shades on the tooth, then it is important to take multiple images for your lab technician. The lighting of the room can also cause a mismatch so be sure to know what lighting you are using.

Custom Shade



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At times you will not be able to accurately get a complex shade on a tooth. In these instances, you will need to send your patient for a custom shade. If you work with a local lab this is possible. If you work with a mail order lab, you will want to work at taking a very good shade and learn dental photography so that you can send your technician high quality photos for them to work from. When you want a custom shade taken, in the Tab# or shade area stipulate it by saying you want a custom shade by your lab.



Written Instructions

RX INSTRUCTIONS _____

Please fabricate an Emax Crown for #9.

See notes for shade, Shade pictures were taken and emailed.

The patient would like to mimic the rounded shape of #8 incisal edge

Thank you

SEE BACK OF RX FOR ADDITIONAL INFORMATION

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In the instructions area restate what you would like done. This is also the area to related special instructions to the lab like a patient wanting to match a specific feature of a neighboring tooth. Also, you can communicate that a certain portion of your margin might be suspect. This is your narrative, be specific on what you want. Your lab can not read your mind and will use the available information that you give them to fabricate the crown. Many times, the reason you do not get back what you wanted is poor communication on the doctor side.

Lab Process



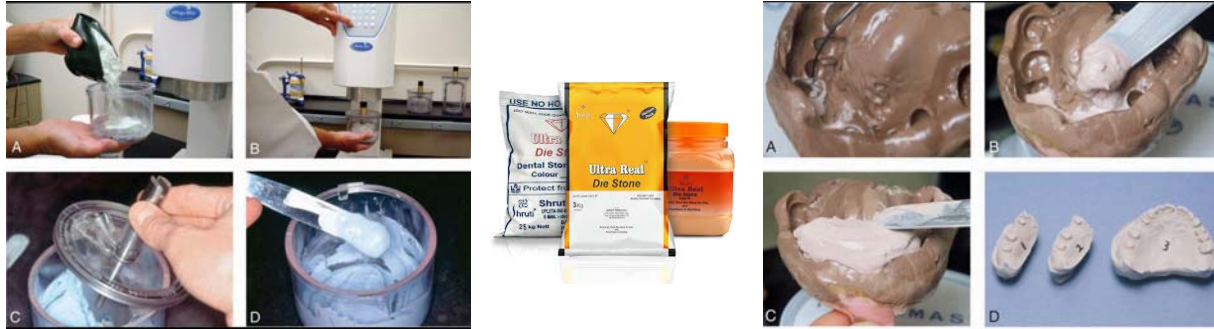
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Over the last decade the process for fabricating a crown has changed a lot. Traditionally, it was completed with stone models and either the lost wax technique and stacking porcelain. Today, labs have at their disposal 3D CAD/CAM as well as 3D printing. We will cover both of these processes so that you understand where problems can come about and how to avoid them.

Traditional Method



Pictures from Contemporary Fixed Prosthodontics 6th Edition

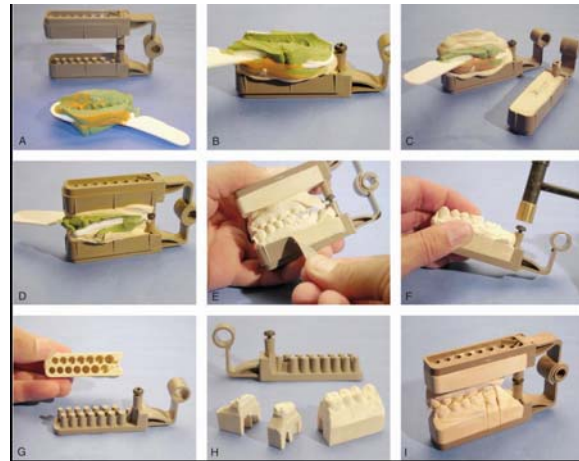
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When you make an analog impression, it is sanitized in your office and then sent to your lab. Most dentists will use a PVS (polyvinyl siloxane) material. This material can be poured up in stone multiple times and will keep its dimensional stability for up to 2 weeks. Once the lab receives your impression they will probably sanitize it again and pour it up in a die stone. These stone materials are generally harder than the traditional yellow stone. Many labs will then pour a second model to keep as an untrimmed dye. This will be explained more later.

Traditional Method



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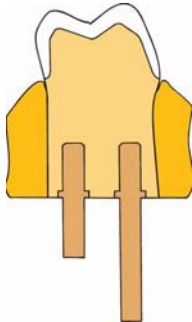
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It generally will take about an hour for the dye stone to cure. It is then removed from the impression and trimmed of the excess stone that is not needed. If you took an impression with a triple tray or double bite tray the impression will be poured and then it can be mounted at the same time. This is demonstrated in the image in the slide.

Pindexing



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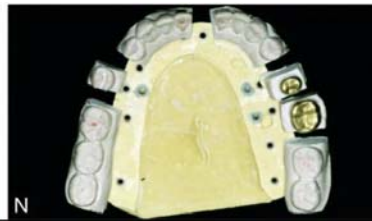
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Pictures from Contemporary Fixed Prosthodontics 6th Edition

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Pindexing is the process of placing pins into your dye that will allow you to remove it from the model to work on the crown and then return it as you need. The pictures in the slide show the process. You will drill small holes into your dye and then glue the pins into the dye. It is important to place pins into the are of the teeth that will be crowned. Sleeves will be placed on the exposed ends of the pins and then your pindexed dye will be placed in stone and allowed to set. Once the stone base is set, you should be able to pull out your complete dye from the base.

Sectioning



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The next step is to section your dye. Using a hand-saw you will section the die into smaller sections. The tooth or teeth that will need crowns will be sectioned into individual sections. This will leave you with multiple sections that can now be removed and reinserted into the stable dye

Mounting



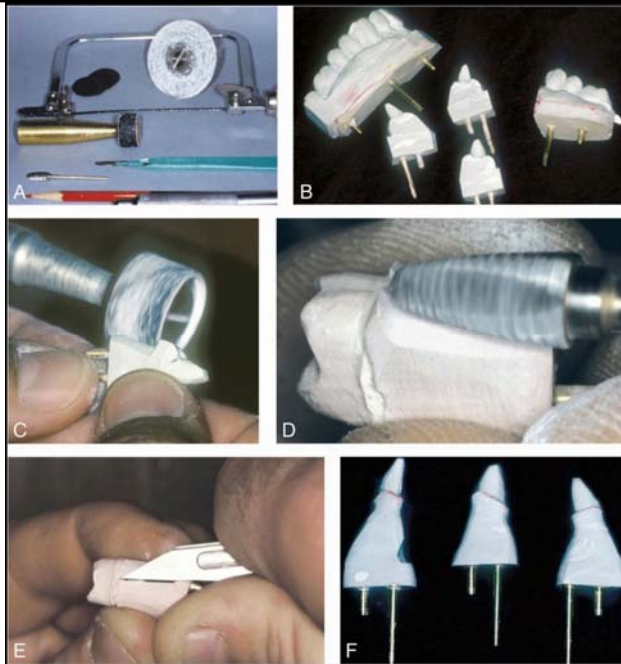
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If your case has not been mounted on an articulator or a hinged appliance (aka snake biter) you will mount it now. This will allow you to more easily check the patient's occlusion on your wax up and final crown.

Dye Trimming



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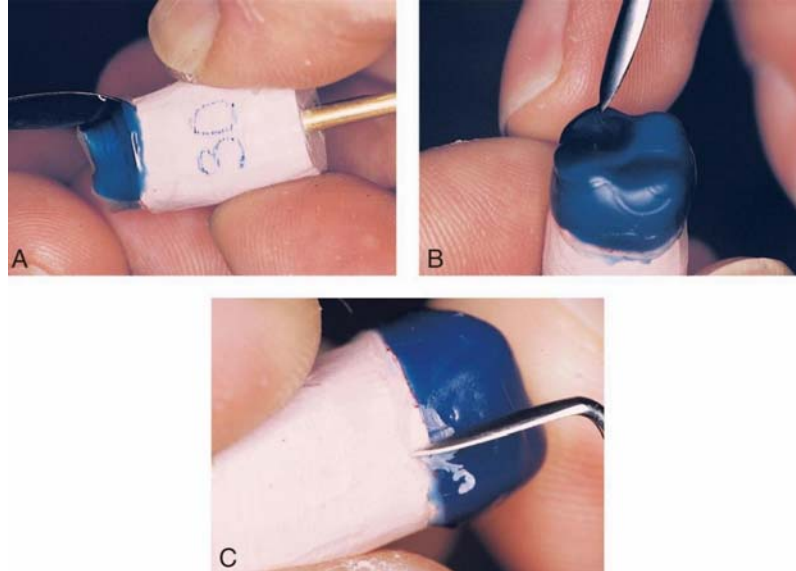
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Pictures from Contemporary Fixed Prosthodontics 6th Edition

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Now that your complete dye is sectioned, you will need to trim the individual dyes of the tooth or teeth that need crowns. Trimming removed the surround gingival tissues and excess to expose your margin for easy waxing. Once the dye has been trimmed you will need to paint on a dye hardener to prevent chipping. Additionally, you will want to paint on dye spacer. This will approximate the film thickness of your luting agent.

Wax up



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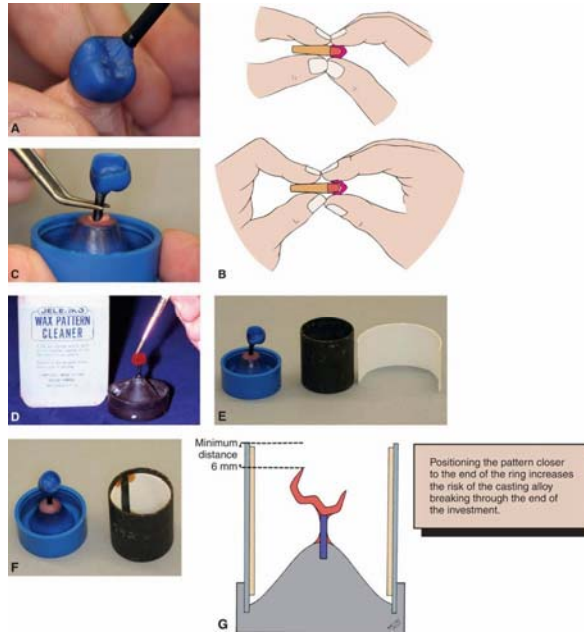
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Now that your dye is trimmed, you will need to do a full contour wax up. If you are doing a full metal crown like a gold crown, you are done and ready to proceed to the next step. If you are doing a porcelain fused to metal crown you will need to cut the wax back to make room for the eventual stacking of porcelain. In this process you will be using the lost wax technique. It is the same if you are making jewelry, and this technique has been used for centuries.

Sprue



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In order to cast your wax up an extension needed to be added, it is called a sprue. The sprue is placed in a large area like the Buccal or lingual or a cusp tip. The sprue will act as a channel for the liquid metal to flow through. You do not want to place it on the occlusal or interproximal surfaces. Doing so will alter these surfaces and may affect your fit after the sprue is removed.

Investing



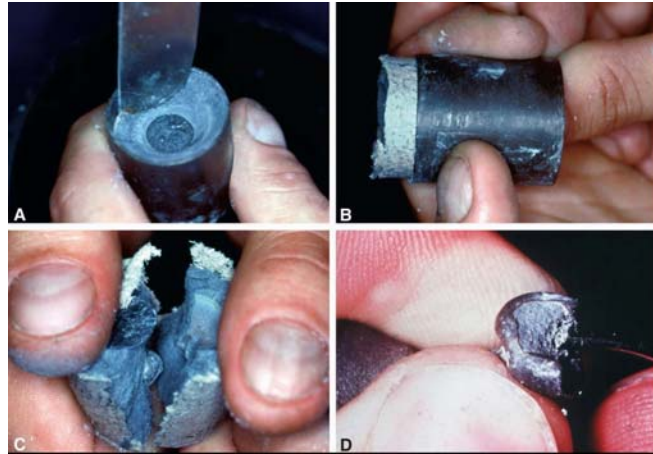
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When you complete your wax up it is time to invest it. An extension called a sprue is added to your crown in a large area like the Buccal or lingual. The sprue will act as a channel for the liquid metal to flow through. You do not want to place it on the occlusal or interproximal surfaces. Doing so will alter these surfaces and may affect your fit after the sprue is removed. The wax-up and sprue and then placed into an investing cylinder and investment stone is pour around it. This is allowed to harden then the ring is placed into the oven at a high temperature to burn out the wax. When this process is complete there will be a void in the investment stone in the shape of the crown with a tunnel leading to the exterior that was the sprue.

Casting



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It is now time to cast your crown. The metal is placed into a crucible and heated with a torch. This metal can then be placed into an injection machine or into a casting ring that will use centripetal force to inject the metal into the investing casting. It is important to get your metal into a liquid state that moves freely, if not you will have voids in your casting. When using the casting ring, it will need to be wound up and then secured. When the metal is ready it is released and allowed to spin. Let the device finish spinning before you try and remove the investment ring. Allow for some cooling time of the metal and then break out the casting. The sprue will need to be removed and the casting evaluated for flaws and voids as well as treated and polished. If you are doing a PFM, it is time to stack porcelain.

Stacking Porcelain



Pictures from Pocket Dentistry



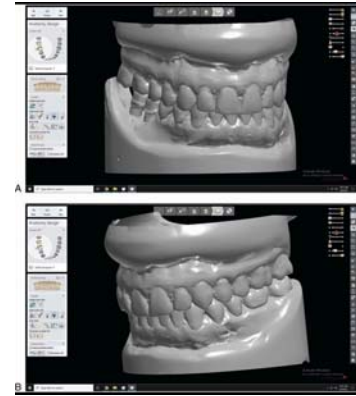
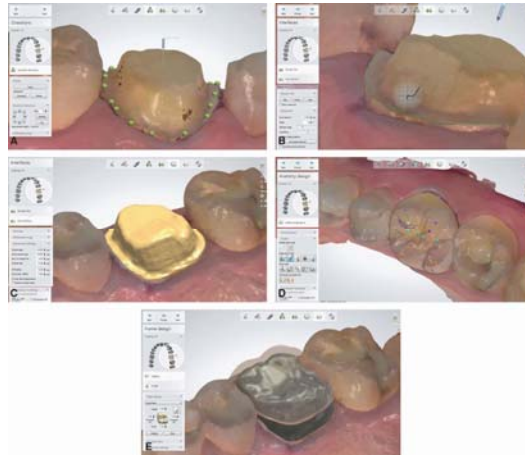
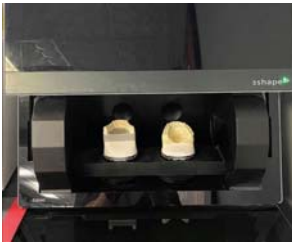
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Porcelain comes in separate liquid and powder. They will need to be mixed and with a brush applied to the restoration in small amounts and then fired in an oven. This process is repeated until the desired shape and size is achieved. The occlusion is checked and adjusted and the restoration is polished and glazed. A good lab will then send their crown to a quality control person that will check the work under magnification to make sure it is ready to be sent to you.

Emax and Zirconia



Pictures from Contemporary Fixed Prosthodontics 6th Edition

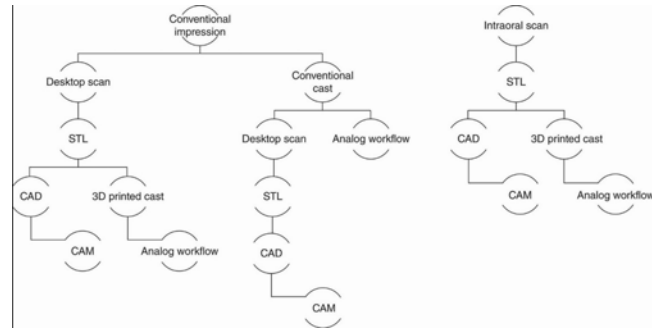
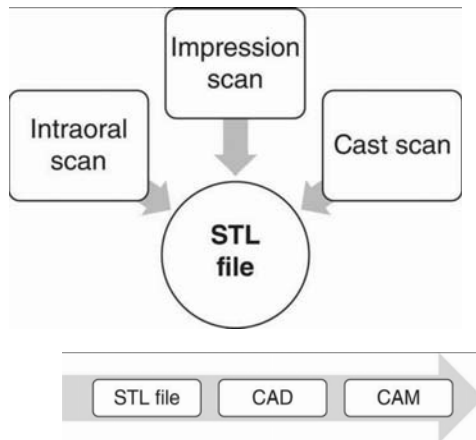
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12
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If you make a traditional impression and want to have an Emax or Zirconia crown created, the lab will either scan the impression and make a virtual model or more like scan the poured dye and design your restoration virtually and mill the crown.

Lab Process with a Scan



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12
1

With the advent of a scanners and the increased use of scans as a final impression, labs have had to adapt to the changing land scape. Scans will come as an STL. File that can easily be emailed to your lab. These files can be uploaded into 3D CAD/CAM software to plan and mill a crown. They also can be used to 3D print a Dye that then can be used to make a crown via the lost wax technique. Many labs will still print a 3D model even if they are going to used CAD/CAM. This is to allow them to check the occlusion and interproximal contacts before they send it back to you. Additionally, it is very handy to have a dye available to help seat the crown own to make adjustments to the crown chairside.